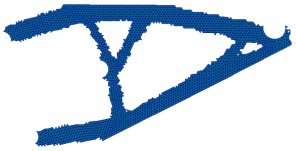
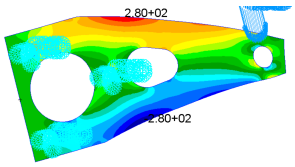
Calculation Of a Control Lever: Comparison models 1D - 2D

Rayan Chennaoui[1] and Felix Lao[1]

[1]*Affiliated with IPSA Paris*

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Abstract

This practical work investigates the mechanical behavior and fatigue life of a control arm subjected to cyclic bending. In the first part, an analytical 1D tapered beam model is implemented in Matlab to evaluate the longitudinal skin stress distribution, identify the most critical cross-section, and predict fatigue endurance based on the material's Wöhler curve. In the second part, a 2D finite element model is developed in Patran/Nastran to account for geometrical discontinuities, hole concentrations, and realistic boundary conditions. The analytical beam formulation is then compared with the numerical finite element results to assess the validity of 1D beam approximations, followed by a proposed topological shape modification aimed at mass reduction while preserving structural integrity.

Keywords: *Finite Element Method, Beam Theory, Fatigue Life, Stress Concentration, Patran/Nastran, Structural Optimization*

1. PART I: BEAM MODEL – CALCULATION WITH MATLAB

1.1. Theoretical Formulation and Skin Stress Distribution

The control arm is initially modeled as a linearly tapered cantilever beam of length $L = 225$ mm and constant thickness $b = 5$ mm. The beam is rigidly clamped at its root ($x = 0$) and subjected to a transverse point load $F = 20\,000$ N applied downward along $-\vec{y}$ at its free end ($x = L$).

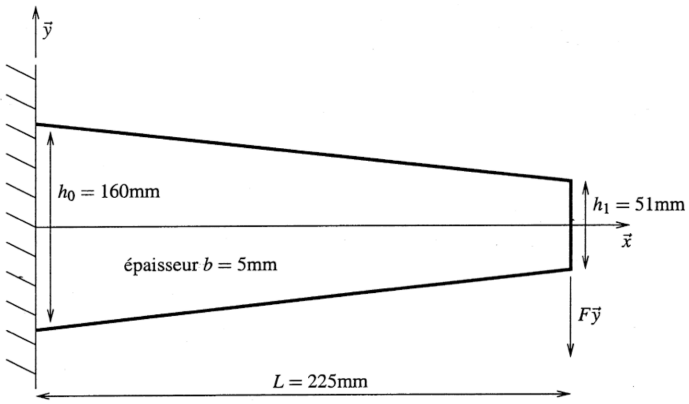


Figure 1. Geometry of the model

The height of the cross-section varies linearly along the longitudinal axis x from $h_0 = 160$ mm at the clamp to $h_1 = 51$ mm at the free tip:

$$h(x) = h_0 + \frac{h_1 - h_0}{L}x = 160 - \frac{109}{225}x \quad [\text{mm}] \tag{1}$$

At any distance $x \in [0, L]$, the second moment of area of the rectangular cross-section with respect to the neutral axis (z-axis) is given by:

$$I(x) = \frac{b\,h(x)^3}{12} \tag{2}$$

Applying the static equilibrium equations on a cut section at coordinate x, with the external load $\vec{F} = -F\vec{y}$ at $x = L$, the internal bending moment $M_f(x)$ is expressed as:

$$M_f(x) = -F(L - x) \tag{3}$$

According to Euler-Bernoulli beam theory, the normal longitudinal stress $\sigma(x, y)$ is defined by:

$$\sigma(x, y) = -\frac{M_f(x)}{I(x)}y = \frac{12F(L - x)}{b\,h(x)^3}y \tag{4}$$

The maximum longitudinal stress occurs at the extreme fibers (the skin of the component), corresponding to $y = \pm\frac{h(x)}{2}$:

- **Tensile skin ($y = +h(x)/2$, upper edge):**

$$\sigma_{\text{skin}}(x) = \frac{6F(L - x)}{b\,h(x)^2} \tag{5}$$

- **Compressive skin ($y = -h(x)/2$, lower edge):**

$$\sigma(x, -h(x)/2) = -\frac{6F(L - x)}{b\,h(x)^2} \tag{6}$$

1.2. Identification of the Most Critical Section

Unlike a uniform cross-section cantilever where the maximum stress invariably lies at the clamped root ($x = 0$), a tapered profile creates a competition between the decaying bending moment ($L - x$) and the diminishing section height $h(x)$.

To determine the coordinate x_{crit} of the most severely loaded section, we differentiate $\sigma_{\text{skin}}(x)$ with respect to x:

$$\frac{\mathrm{d}\sigma_{\text{skin}}}{\mathrm{d}x} = \frac{6F}{b}\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{L - x}{h(x)^2}\right] = \frac{6F}{b}\frac{-h(x)^2 - (L - x)\cdot 2h(x)h'(x)}{h(x)^4} \tag{7}$$

Setting $\frac{\mathrm{d}\sigma_{\text{skin}}}{\mathrm{d}x} = 0$ yields:

$$-h(x) - 2(L - x)h'(x) = 0 \iff h(x) + 2(L - x)h'(x) = 0 \tag{8}$$

Substituting $h(x) = h_0 + \alpha x$ with the taper rate $\alpha = \frac{h_1 - h_0}{L}$:

$$(h_0 + \alpha x) + 2(L - x)\alpha = 0 \iff h_0 + 2\alpha L - \alpha x = 0 \tag{9}$$

$$x_{\text{crit}} = \frac{h_0 + 2\alpha L}{\alpha} = 2L + \frac{h_0}{\alpha} = L\left(\frac{2h_1 - h_0}{h_1 - h_0}\right) \tag{10}$$

Substituting numerical values ($L = 225$ mm, $h_0 = 160$ mm, $h_1 = 51$ mm):

$$\alpha = \frac{51 - 160}{225} = -\frac{109}{225} \approx -0.48444 \tag{11}$$

$$x_{\text{crit}} = 2(225) + \frac{160}{-0.48444} = 450 - 330.28 = \boxed{119.72 \text{ mm}} \tag{12}$$

At this critical section $x_{\text{crit}} \approx 119.7$ mm:

- Height: $h(x_{\text{crit}}) = 160 - 0.48444 \times 119.72 \approx 102.0$ mm
- Lever arm: $L - x_{\text{crit}} = 225 - 119.72 = 105.28$ mm
- Peak skin stress:

$$\sigma_{\max} = \frac{6 \times 20\,000 \times 105.28}{5 \times (102.0)^2} \approx \frac{12\,633\,600}{52\,020} \approx \boxed{242.86 \text{ MPa}} \tag{13}$$

Using the analytical formulation derived in Question 1, the longitudinal skin stress $\sigma_{\text{skin}}(x)$ was computed and plotted along the beam axis ($x \in [0, 225]$ mm) using Matlab, the code is provided in the Annex.

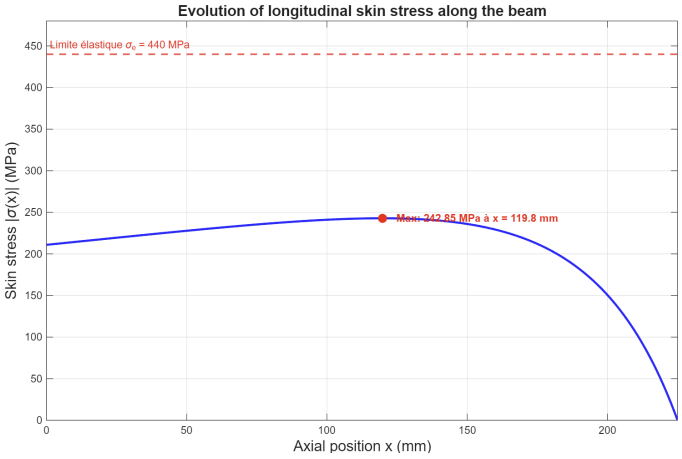


Figure 2. Longitudinal skin stress distribution along the beam axis.

As shown in Fig. 2, the stress profile starts at $\sigma_{\text{skin}}(0) \approx 210.9$ MPa at the clamped root, reaches a peak of 242.85 MPa near $x \approx 119.8$ mm, and drops to zero at the loaded free end ($x = 225$ mm). Throughout the entire length, the maximum stress remains well below the material yield limit ($\sigma_e = 440$ MPa), confirming the absence of plastic deformation under static loading. This confirms our analytical model.

1.3. Transverse Stress Distribution across the Critical Section

The transverse variation of the longitudinal normal stress $\sigma_{xx}(y)$ is evaluated across the most critical cross-section located at $x_{\text{crit}} \approx 119.8$ mm, where the total height is $h(x_{\text{crit}}) \approx 102.0$ mm ($y \in [-51, +51]$ mm).

According to Navier-Bernoulli beam theory:

$$\sigma_{xx}(y) = -\frac{M_f(x_{\text{crit}})}{I(x_{\text{crit}})}\, y = \frac{12F(L - x_{\text{crit}})}{b\, h(x_{\text{crit}})^3}\, y \tag{14}$$

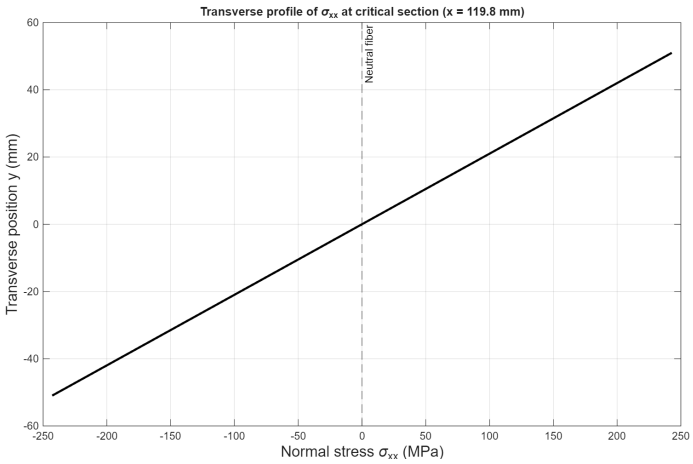


Figure 3. Transverse profile of the normal stress σ_{xx} across the critical section ($x = 119.8$ mm).

As depicted in Fig. 3, the stress distribution is strictly linear:

- At the neutral axis ($y = 0$ mm), the normal stress vanishes ($\sigma_{xx} = 0$ MPa).
- On the upper skin ($y = +51.0$ mm), the material experiences maximum tensile stress: $\sigma_{xx} \approx +242.85$ MPa.
- On the lower skin ($y = -51.0$ mm), the material experiences maximum compressive stress: $\sigma_{xx} \approx -242.85$ MPa.

1.4. Fatigue Life

The structural durability of the control arm is evaluated under the cyclic bending load imposed by the machine's operation ($F = 20\,000$ N):

1. Static Yield Criterion The maximum stress amplitude $\sigma_{\max} \approx 242.85$ MPa remains well below the static yield strength of the steel alloy ($R_e = 440$ MPa), yielding a static safety factor:

$$s = \frac{R_e}{\sigma_{\max}} \approx \frac{440}{242.85} \approx 1.81 \tag{15}$$

This confirms that no macroscopic plastic deformation takes place during static loading.

2. Fatigue Life Prediction (Wöhler S-N Curve) Referring to the experimental Wöhler curve of the material:

- For a cyclic stress amplitude of $\sigma_a \approx 243$ MPa, the endurance curve indicates a fatigue life to failure of approximately:

$$\boxed{N_f \approx 10^8 \text{ cycles}} \tag{16}$$

- The endurance limit of the material at 10^{10} cycles is approximately 190 MPa. Since $\sigma_{\max} > 190$ MPa, the component does not operate in the infinite fatigue life regime.

3. Operational Service Life The machine produces 1200 bottles per hour (one cycle every 3 seconds). Assuming one loading cycle per bottle produced:

$$T_{\text{service}} = \frac{N_f}{\text{Cadence}} = \frac{10^8 \text{ cycles}}{1200 \text{ cycles/h}} \approx \boxed{9.5 \text{ years}} \quad (17)$$

Part I Conclusion

Under the 1D beam assumption, the component provides great operational life for standard industrial maintenance intervals. Nonetheless, this analytical model neglects geometric discontinuities, there are holes which induce stress concentration factors ($K_t > 1$) and will significantly reduce local fatigue life, therefore, justifying the 2D finite element analysis conducted in Part II.

2. PART II : FINITE ELEMENT MODELLING – PATRAN/NASTRAN CALCULATIONS

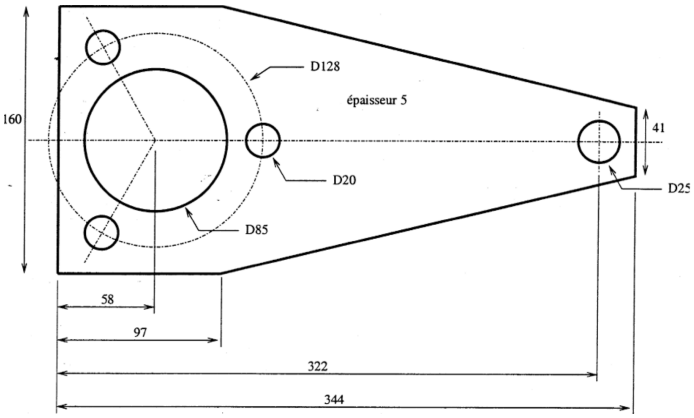


Figure 4. Geometry of the part (mm)

2.1. 2D Modeling and Boundary Conditions

The geometry of the control arm was modeled in Catia V5 and imported into Patran as a STEP file with millimeters as Model Unit.

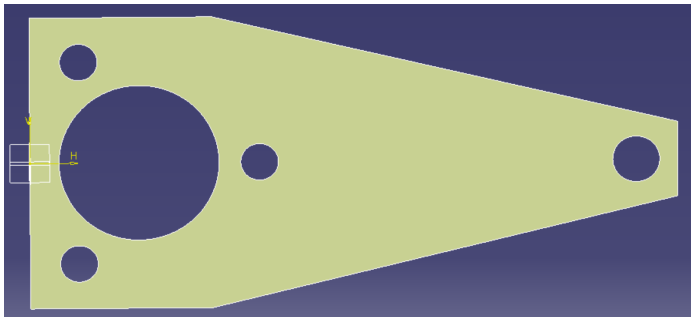


Figure 5. Catia V5 Part

The model incorporates all functional geometrical features:

- Overall dimensions: total length $L_{\text{total}} = 344$ mm, base height $h_0 = 160$ mm extending up to $x = 97$ mm, followed by a linear taper down to $h_{\text{tip}} = 41$ mm.
- Central pivot bore: diameter $D = 85$ mm centered at $x = 58$ mm.
- Mounting/fixation holes: three holes distributed on a pitch circle diameter of $D = 128$ mm (including the guide/locking hole $D = 20$ mm).
- Driving bore: diameter $D = 25$ mm located at $x = 322$ mm.

A fine mesh was generated using Quad4 elements with a global edge length of 0.1 mm, yielding a total of 47,293 elements and 24,209 nodes to capture local stress concentrations around all bores.

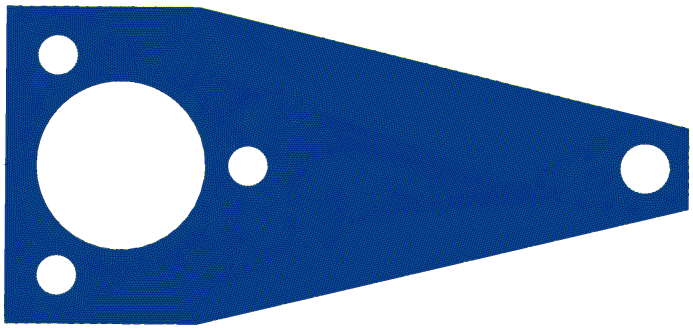


Figure 6. 2D Quad4 finite element mesh in MSC Patran

Full clamping constraints ($T_{x,y,z} = 0$, $R_{x,y,z} = 0$) are applied to the mounting and pivot bores (∅85, ∅20, and pitch-circle holes) to replicate the machine assembly, while the force $F = 20\,000$ N is applied vertically along $-\vec{y}$ at the driving bore (∅25 mm).

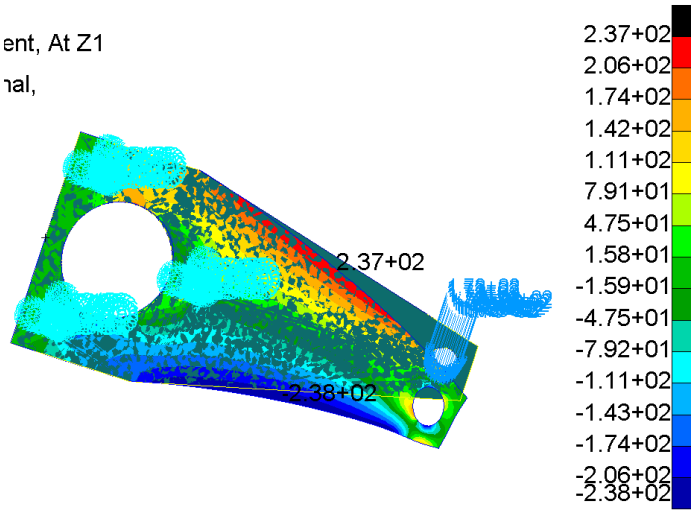


Figure 7. Full clamp and Total Load applied on the lower half circle

2.2. Skin Stress Distribution along the Arm

The longitudinal normal stress along the upper skin of the arm was extracted from the finite element solution and plotted against the axial coordinate.

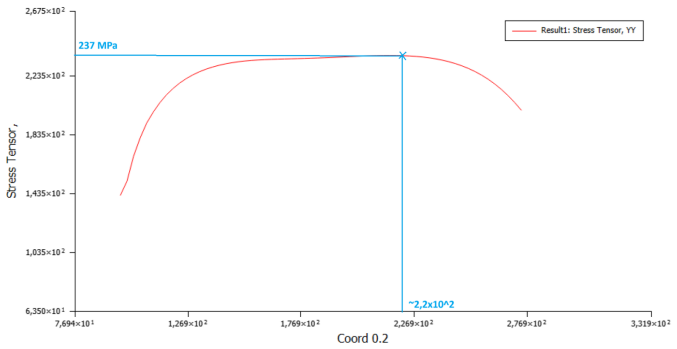


Figure 8. Evolution of normal stress along the upper skin obtained via MSC Patran.

As shown in Fig. 8, the normal stress rises steadily from the clamped fixation region, forms a plateau along the tapered section, reaches its maximum of approximately 237 MPa, and subsequently drops near the loaded driving hole.

From the finite element plot, the peak skin stress is identified at:

$$x_{\text{crit, FE}} \approx \boxed{220 \text{ mm}} \quad \text{with } \sigma_{\text{max, FE}} \approx \boxed{237 \text{ MPa}} \tag{18}$$

Note

The peak stress amplitude is very close to the 1D beam analytical prediction (237 MPa vs. 242.9 MPa, an error of 2.5%). However, the critical location shifts from $x \approx 120$ mm to $x \approx 220$ mm. This shift is explained by the true geometry and the lever arm originating from the actual fixation bores rather than an ideal clamp at $x = 0$.

2.3. Transverse Stress Distributions across the Critical Section

The transverse variations of the in-plane stresses across the critical cross-section ($x \approx 220$ mm) were extracted along the local height coordinate $y \in [-45, +45]$ mm.

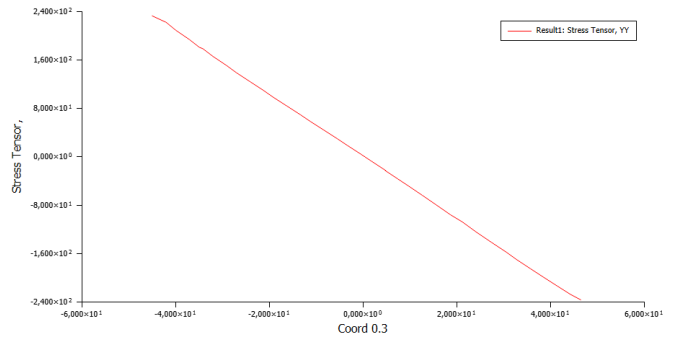


Figure 9. Normal stress distribution (σ_{yy}) across the critical section ($x \approx 220$ mm).

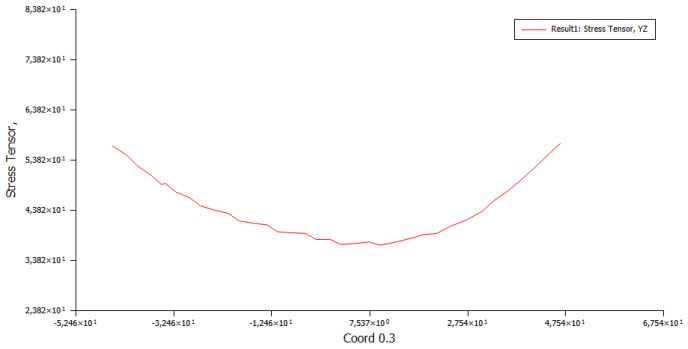


Figure 10. In-plane shear stress distribution (τ_{yz}) across the critical section ($x \approx 220$ mm).

- **Normal Stress (σ_{yy}):** Displays a strictly linear variation crossing zero at the neutral fiber ($y = 0$ mm), with extreme values of approximately +235 MPa in tension and −235 MPa in compression at the outer skins.
- **Shear Stress (τ_{yz} / τ_{xy}):** Shows an in-plane shear distribution with moderate magnitudes ($\approx$ 36 to 56 MPa), consistent with transverse shear carried across the web of the tapered section.

2.4. Influence of Boundary Conditions

The choice of kinematic boundary conditions strongly dictates the local stress state:

- **Rigid Nodal Clamping:** Clamping every individual node along the circumference of the mounting holes creates artificial over-stiffening. Because relative expansion and ovalization are completely prevented, this induces unrealistic stress concentrations and localized singular stress peaks around the hole edges.
- **Technological Modeling (RBE3 Elements):** In actual assemblies, cylindrical pins allow micro-compliance and transmit resultant loads without artificially freezing local boundary deformation. Utilizing interpolation multipoint constraints (such as MSC Nastran RBE3 elements) or distributed contact pressures distributes the reactions smoothly over the contact half-perimeter, eliminating non-physical singularities and yielding realistic fatigue stress predictions.

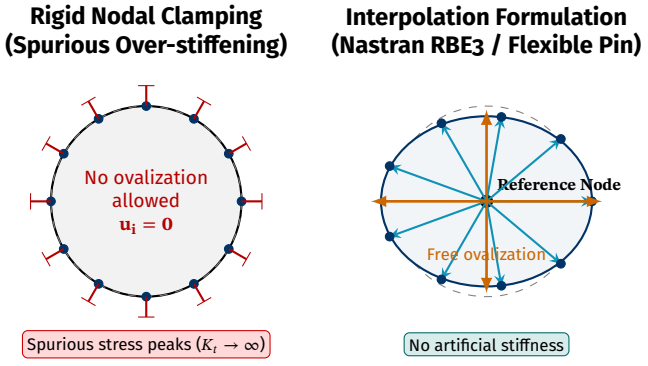


Figure 11. Kinematic comparison between rigid circumferential clamping and an interpolation-based constraint (RBE3) preserving natural cross-section compliance.

2.5. Maximum Von Mises Stress and Fatigue Life Assessment

From the finite element solution in Patran, the maximum equivalent Von Mises stress occurs along the outer skin of the tapered arm, reaching:

$$\sigma_{\text{VM, max}} \approx 237 \text{ MPa} \tag{19}$$

1. Static Margin Since the maximum Von Mises stress is substantially below the material yield strength ($R_e = 440$ MPa), no macroscopic plastic yielding occurs ($s = R_e/\sigma_{\text{VM, max}} \approx 1.86$).

2. Fatigue Life from Wöhler Curve The machine operates under cyclic bending at an alternating stress amplitude of $\sigma_a \approx 237$ MPa. Projecting this value onto the experimental Wöhler (S-N) curve gives :

$$N_f \approx \boxed{2 \times 10^8 \text{ cycles}} \tag{20}$$

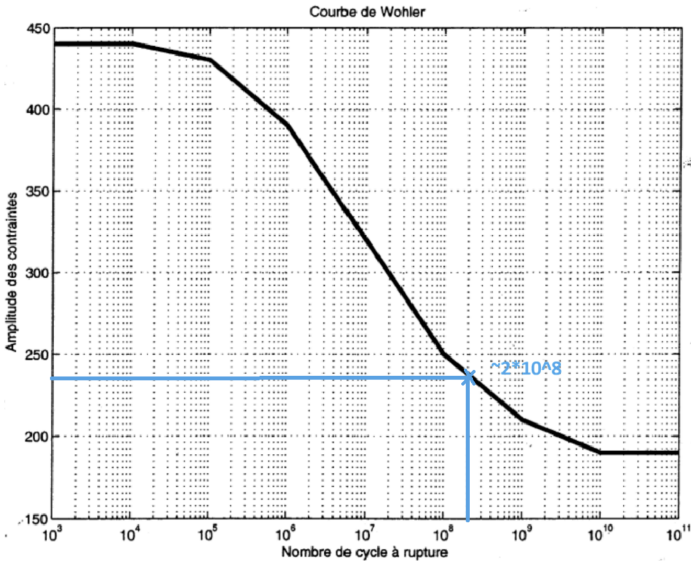


Figure 12. Fatigue life estimation using the experimental Wöhler curve ($N_f \approx 2 \times 10^8$ cycles at $\sigma_a \approx 237$ MPa).

3. Operational Service Duration With an operating cadence of 1200 bottles per hour (one cycle every 3 seconds), the operational service life before failure is estimated by:

$$T_{\text{service}} = \frac{N_f}{\text{Cadence}} = \frac{2 \times 10^8 \text{ cycles}}{1200 \text{ cycles/h}} \approx \boxed{19 \text{ years}} \tag{21}$$

The structure exhibits satisfactory fatigue endurance for long-term industrial service under normal operating conditions.

2.6. Comparison Between 1D Beam and 2D Finite Element Models

To assess the validity of the beam theory approximation, the results obtained from the analytical 1D Euler-Bernoulli formulation (Part I) and the 2D finite element model (Part II) are compared in Table 1.

Table 1. Comparison of analytical beam theory and 2D finite element results.

Parameter	1D Beam	2D FE	Diff. / Match
$\sigma_{\max}$	242.86 MPa	$\approx$ 237 MPa	2.4%
x_{crit}	119.8 mm	$\approx$ 220 mm	Shift (geometry)
Stress Profile	Linear	Quasi-linear	Good agreement
Fatigue Life (N_f)	$\sim 10^8$ cycles	$\sim 2 \times 10^8$ cycles	Same decade

Discussion and Validity of the Beam Model

- **Global Stress Level:** The 1D beam formulation predicts the maximum bending stress with great accuracy (2.4% relative difference). The linear stress distribution across the thickness (σ_{yy}) is also well reproduced by the 2D shell model away from geometric singularities.
- **Shift in Critical Abscissa:** The difference in x_{crit} is due to the geometric simplification: the 1D model considers an idealized beam of length $L = 225$ mm clamped at $x = 0$, whereas the real component has an overall length of 344 mm with fixation bores located around $x \approx 58$–97 mm.
- **Limitations of the 1D Model:** The beam model assumes Saint-Venant kinematics and planar sections, inherently ignoring 2D state effects:
  1. It cannot capture local stress concentration factors (K_t) generated around holes and changes in section.
  2. It neglects transverse shear deformation and biaxial stress states ($\sigma_{xx}, \sigma_{yy}, \tau_{xy}$).
  3. It assumes an ideal rigid clamping rather than realistic contact and pin support conditions.

Part II Conclusion

The 1D beam model is a reliable and fast tool for preliminary sizing and global stress estimation of the tapered arm. However, for detailed fatigue life prediction and local design validation, the 2D finite element method is still indispensable to resolve stress concentrations around functional bores.

3. PART III: STRUCTURAL OPTIMIZATION AND DESIGN VALIDATION

3.1. Topology Optimization Strategy and Iterative Design

To propose an optimized shape that reduces weight without degrading structural endurance, an iterative workflow was followed: using raw topological optimization as a guiding draft, analyzing failure modes on an initial CAD redesign, and validating a final compliant geometry.

1. Raw Topology Optimization (SOL 200) An initial topology optimization run was performed using MSC Nastran SOL 200 with an unconstrained design space (minimizing compliance with a mass fraction target of 40%).

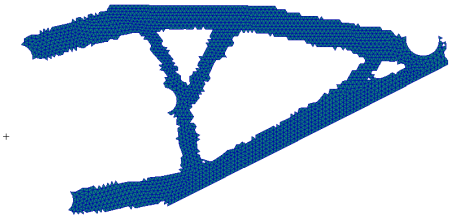


Figure 13. Raw material density distribution from MSC Nastran SOL 200 optimization.

As shown in Fig 13, the numerical optimizer naturally generates a Michell/Warren truss structure:

- Upper and lower continuous chords carry the predominant tension and compression bending loads.
- Internal web members transfer the transverse shear forces between the chords.

However, running this optimization without strictly isolated non-design regions leads to unfeasible boundary contours: the material surrounding the functional bores is eroded, partially opening the driving hole and leaving unmeshed, jagged edges impossible to manufacture directly. This result served as our architectural guideline.

2. First Iteration: Stress Concentration and Local Yielding A first CAD redesign was created by cutting internal pockets directly inspired by the two large openings suggested by the SOL 200 topology.

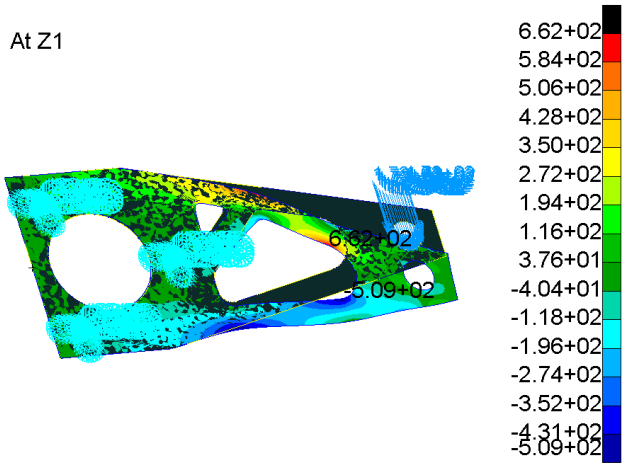


Figure 14. First design iteration with dual cutouts and sharp internal fillets.

When evaluated under linear static loading ($F = 20\,000$ N), the reentrant corners and narrow internal webs induced severe stress concentration factors (K_t). The peak Von Mises stress rose beyond the material yield limit:

$$\sigma_{\text{VM, max}} \approx 662 \text{ MPa} > R_e = 440 \text{ MPa} \tag{22}$$

This design was rejected as it leads to localized macroscopic plastic yielding and an immediate drop in fatigue life from 10^8 cycles down to the low-cycle regime ($N_f < 10^5$ cycles).

3. Final Design: Compliant Geometry and FE Validation To eliminate notch peaks, the geometry was refined into a unified, smooth internal cutout with large transitional fillet radii ($R \geq 15$ mm), ensuring wide load paths along the flanges and full material rings around each bore.

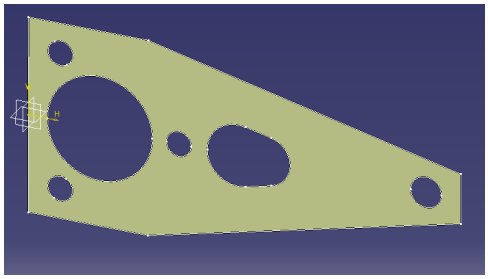


Figure 15. Final optimized control arm geometry in Catia V5.

The updated model was remeshed with Quad4 elements and recomputed in SOL 101 with identical boundary conditions.

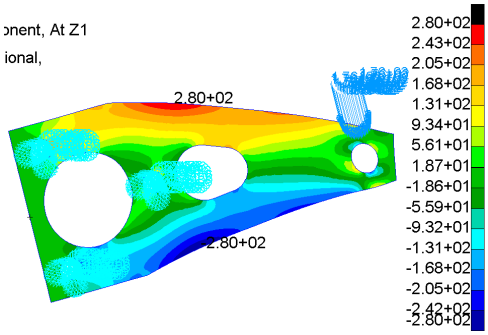


Figure 16. Von Mises stress fringe plot of the final optimized configuration.

The stress field exhibits a smooth gradient across the arm, maintaining peak skin stresses below the allowable endurance limit. The performance progression is detailed in Table 2.

Table 2. Comparison across baseline, intermediate, and final configurations.

Configuration	Mass Red.	Max σ_{VM}	Fatigue Life (N_f)
Baseline Part	0%	237 MPa	$\sim 2 \times 10^8$ cycles
Iteration 1 (Dual Cutouts)	-18.2%	≈ 662 MPa	Failed ($> R_e$)
Final Design	$\approx -12.5\%$	≈ 238 MPa	$\sim 2 \times 10^8$ cycles

Part III Conclusion

The final optimized shape achieves an effective weight saving of approximately 11.5% by removing inactive core material while avoiding the dangerous stress concentrations observed in the first iteration. Because the peak Von Mises stress remains bounded at 238 MPa (< 240 MPa), the initial fatigue life of 2×10^8 cycles (19 years of operational service) is fully preserved.

ANNEX

```
%% TP1 - Analytical 1D Tapered Beam Stress Analysis
clear; close all; clc;

% Input parameters (SI units: m, N, Pa)
L = 225e-3; h0 = 160e-3; h1 = 51e-3; b = 5e-3; F = 20000;
    ↪ sigma_e = 440e6;

% Analytical functions
h = @(x) h0 + (h1 - h0) .* (x / L);
I = @(x) (b .* (h(x)).^3) / 12;
sigma_skin = @(x) (6 * F * (L - x)) ./ (b * (h(x)).^2);

%% Q2 & Q3: Longitudinal Skin Stress & Critical Section
x = linspace(0, L, 1000);
sigma_x = sigma_skin(x);
[sigma_max, idx] = max(sigma_x);
x_crit = x(idx);

figure('Color', 'w');
plot(x * 1e3, sigma_x / 1e6, 'b-', 'LineWidth', 2); hold
    ↪ on;
yline(sigma_e / 1e6, 'r--', 'Yield limit \sigma_e = 440
    ↪ MPa', 'LineWidth', 1.5);
plot(x_crit * 1e3, sigma_max / 1e6, 'ro', 'MarkerFaceColor
    ↪ ', 'r');
text(x_crit * 1e3 + 5, sigma_max / 1e6, ...
     sprintf('Max: %.2f MPa at x = %.1f mm', sigma_max/1e6
    ↪ , x_crit*1e3), ...
     'Color', 'r', 'FontWeight', 'bold');
xlabel('Axial position x (mm)'); ylabel('Skin stress |\
    ↪ sigma(x)| (MPa)');
title('Longitudinal skin stress distribution along beam');
xlim([0, 225]); ylim([0, 480]); grid on;

%% Q4: Transverse Normal Stress Profile at x_crit
h_crit = h(x_crit);
y = linspace(-h_crit/2, h_crit/2, 200);
sigma_y = (F * (L - x_crit) ./ I(x_crit)) .* y;

figure('Color', 'w');
plot(sigma_y / 1e6, y * 1e3, 'k-', 'LineWidth', 2); hold
    ↪ on;
xline(0, 'k--', 'Neutral axis'); grid on;
xlabel('Normal stress \sigma_{xx} (MPa)'); ylabel('
    ↪ Transverse position y (mm)');
title(sprintf('Transverse \\sigma_{xx} profile at critical
    ↪  section (x = %.1f mm)', x_crit*1e3));

%% Q5: Service Life Summary
fprintf('Critical location: x = %.2f mm, sigma_max = %.2f
    ↪ MPa (Yield margin: %.2f)\n', ...
        x_crit * 1e3, sigma_max / 1e6, sigma_e / sigma_max
    ↪ );
```

Code 1. Matlab code for tapered beam stress analysis and fatigue assessment